

Paul Edelman

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Professional Summary

Software engineer pursuing dual degrees in Computer Science and Applied Mathematics at the University of Delaware (Anticipated Graduation: May 2027). Strong foundation in data structures and algorithms (300+ problems solved in C++), with hands-on project experience spanning distributed systems, full-stack web development, machine learning, and Unix/Linux systems programming.

Education

Bachelor of Science in Computer Science, Bachelor of Arts in Applied Mathematics

University of Delaware, Newark, DE

Aug 2023 – May 2027 — Honors Student, GPA: 3.7 — Anticipated Graduation: May 2027

Experience

Software Engineer — DELJIS, Dover, DE

Jun 2026 – Aug 2026 — C#, Podman, systemd, Linux, Kubernetes-style Orchestration, GitLab CI/CD

- Architected a Kubernetes-inspired container orchestration platform in C# on Linux, using Podman to drive the full container lifecycle (provisioning, networking, teardown) from declarative configuration plans.
- Used systemd to supervise each managed container, restarting it automatically according to a configurable restart policy whenever the container's image or configuration changed.
- Detected configuration and image changes via FileSystemWatcher for locally stored configs and WebSocket listeners for configs on a remote server, feeding both paths into the same systemd-driven restart pipeline; container logs and lifecycle events were likewise available locally or streamed to the server over WebSockets for real-time visibility.
- Engineered a REST API as the orchestration backbone, plus a developer-facing CLI for plan registration, deployment control, health monitoring, and local-plan dev mode.
- Automated hotfix delivery via declarative plan updates instead of code redeploys, cutting deployment time approximately 90%.

Software/Data Engineering Intern — AirGreen, New Castle, DE

May 2025 – Sep 2025 — Python, SQL, PostgreSQL

- Developed and optimized Python data pipelines for ingesting, cleaning, and storing real-time IoT sensor data at scale.
- Implemented SQL/PostgreSQL schema optimizations and indexing, reducing query latency by 40% in high-volume analytics workloads.

Undergraduate Teaching Assistant — University of Delaware, Newark, DE

Aug 2024 – May 2025 — Python, Data Structures & Algorithms, Debugging

- Led lab sessions and office hours covering data structures, algorithms, and systems-level debugging, giving and receiving feedback with students in real time.
- Collaborated with faculty to align curriculum with industry software engineering practices.

Projects

Ticker

Go, Distributed Systems, Testing — Jul 2026 – Present

- Building a fault-tolerant, ordered market data distribution system in Go, guaranteeing correct message ordering and continued availability under node failure.
- Implemented the Raft consensus protocol from scratch (leader election, log replication, crash recovery), including test suites to verify correctness under simulated node failures.

Real-time Collaborative Whiteboard

React, JavaScript, WebSockets, Node.js, MongoDB — Oct 2025 – Jan 2026

- Built a full-stack real-time web application with a React frontend and Node.js/Express backend, using WebSockets for multi-user canvas synchronization with sub-100ms latency.
- Designed a MongoDB schema and backend architecture to persist data and maintain state consistency across distributed client sessions.

PredictSports

Python, MySQL, Playwright, TensorFlow/PyTorch — Jan 2025 – Present

- Built an end-to-end sports analytics pipeline: scraped and cleaned decades of historical NFL game data with Playwright, storing it in a structured MySQL database for efficient querying.
- Trained a neural network to forecast game outcomes from the processed dataset, gaining hands-on experience with the full machine learning pipeline from raw data to trained model.

Technical Skills

Languages: Python, JavaScript, C#, Go, C/C++, Java, SQL

Core CS & Problem Solving: Data Structures & Algorithms (300+ problems solved in C++), Distributed Systems, Object-Oriented Programming

Web: React, Node.js/Express, REST API, WebSockets, HTML/CSS

Data & ML: MySQL, PostgreSQL, MongoDB, TensorFlow, PyTorch, pandas, NumPy

Systems & Tools: Linux/UNIX, Podman, Docker, systemd, Kubernetes, Git, GitHub, GitLab, GitLab CI/CD, Debugging, Automated Testing

Modern Workflow: AI-assisted development (Claude Code) for accelerated iteration and debugging